



Ensembles of Classifiers

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The Wisdom of Crowds

nature

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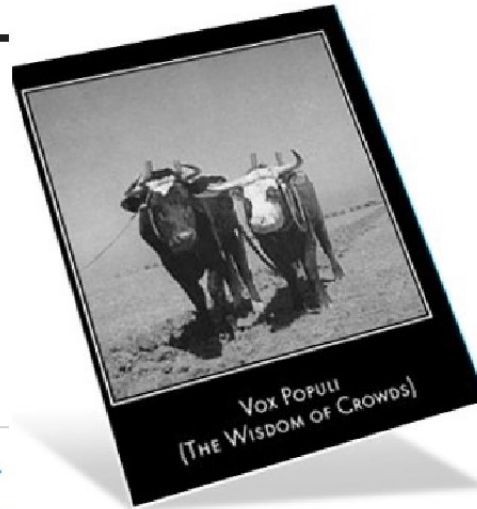
Vox Populi

FRANCIS GALTON

Nature **75**, 450–451(1907) | [Cite this article](#)

Abstract

IN these democratic days, any investigation into the the trustworthiness and peculiarities of popular judgments is of interest. The material about to be discussed refers to a small matter, but is much to the point.





Ask the Audience

► In Wikipedia's current logo, depicting a spherical jigsaw puzzle, which letter is seen on the piece immediately to the right of the one bearing an "Ω"?

A. W

B. X

C. Y

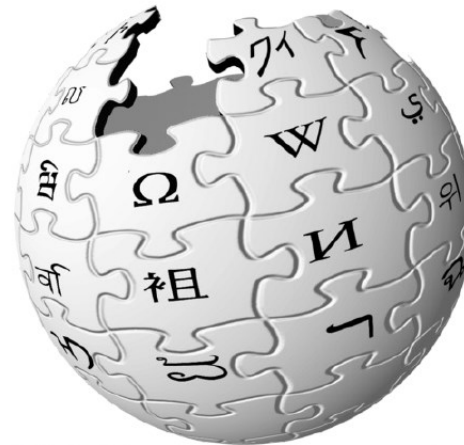
D. Z



Ask the Audience (cont.)

► In Wikipedia's current logo, depicting a spherical jigsaw puzzle, which letter is seen on the piece immediately to the right of the one bearing an "Ω"?

- A. W
- B. X
- C. Y
- D. Z



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Ensemble of Classifiers

► Goal

- Improve accuracy of supervised learning task

► Approach

- Use an ensemble of classifiers, rather than just one

► Challenges

- How to construct ensemble
- How to use individual hypotheses of ensemble to produce a classification



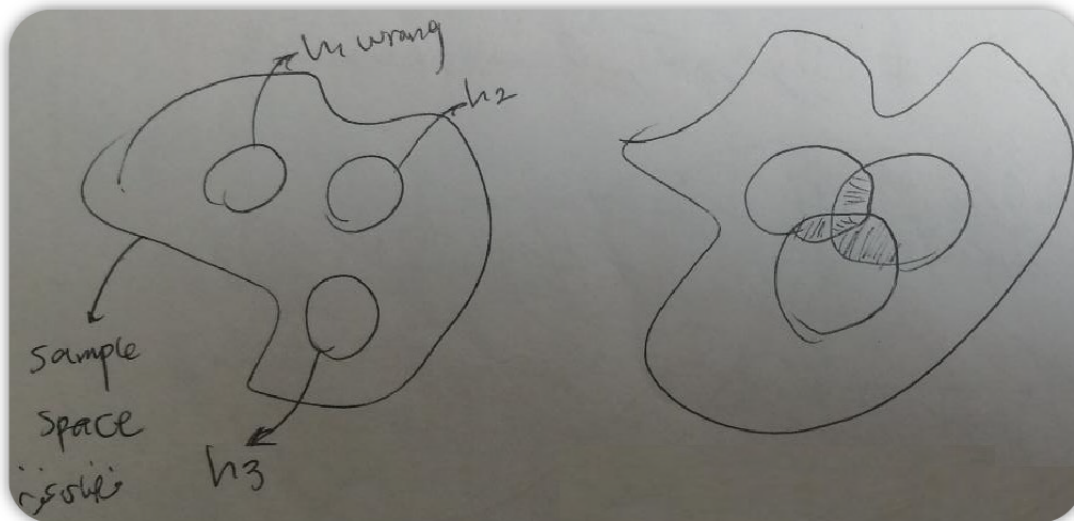
Ensemble of Classifiers (cont.)

- ▶ Given ensemble of L classifiers $h_1, h_2, \dots, h_L$ (called base classifiers)
- ▶ Decision based on combination of h_l 's ($l = 1..L$)
 - ▶ E.g., weighted or unweighted voting
- ▶ How to construct ensemble whose accuracy is better than any individual classifier?



Ensemble of Classifiers (cont.)

- ▶ Combining classifiers by an unweighted voting schema:

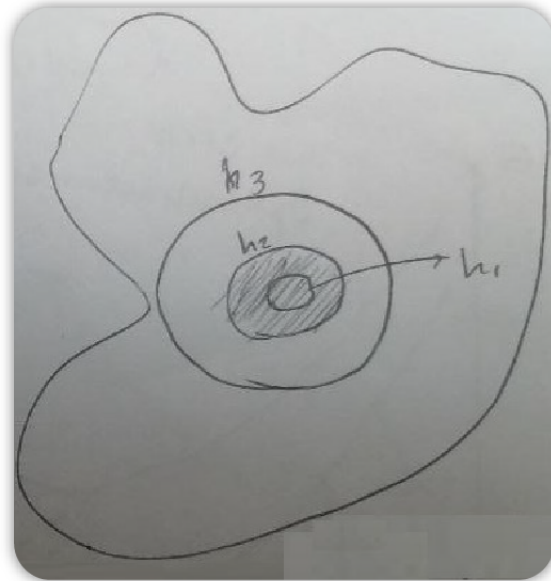


$$H(x) = \text{vote}(h_1(x), h_2(x), h_3(x))$$



Ensemble of Classifiers (cont.)

► Note that h_1 by itself outperforms the ensemble of h_1 , h_2 and h_3 ! how come?





Ensemble of Classifiers (cont.)

- ▶ Ensemble requirements
 - ▶ Individual classifiers disagree
 - ▶ Each classifier's error < 0.5
 - ▶ Weak classifier
 - ▶ Classifiers' errors uncorrelated
- ▶ Then, ensemble will outperform any h_1



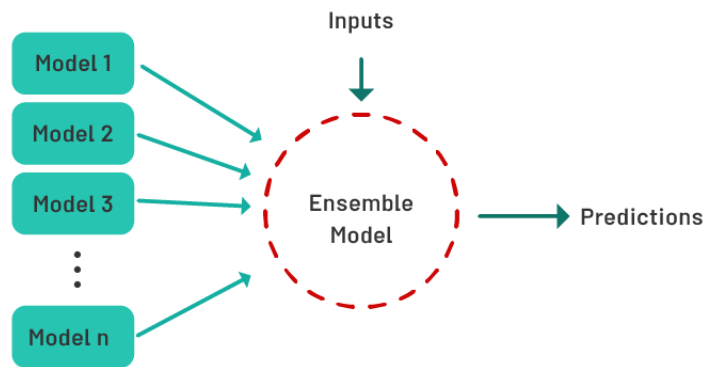
Ensemble Methods

- ▶ Two main categories:
 - I. Bagging (bootstrap aggregation)
 - II. Boosting → Adaboost
- ▶ Combining a set of heterogeneous classifiers: Stacking, blending, voting, ...
- ▶ Graph-based cross-validated committees ensembles
- ...

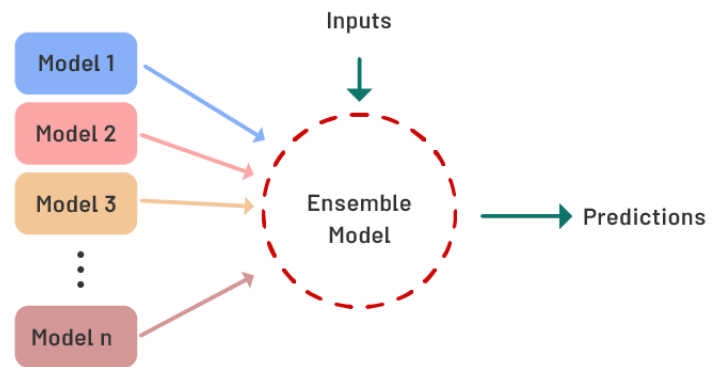


Ensemble Learning

Homogeneous



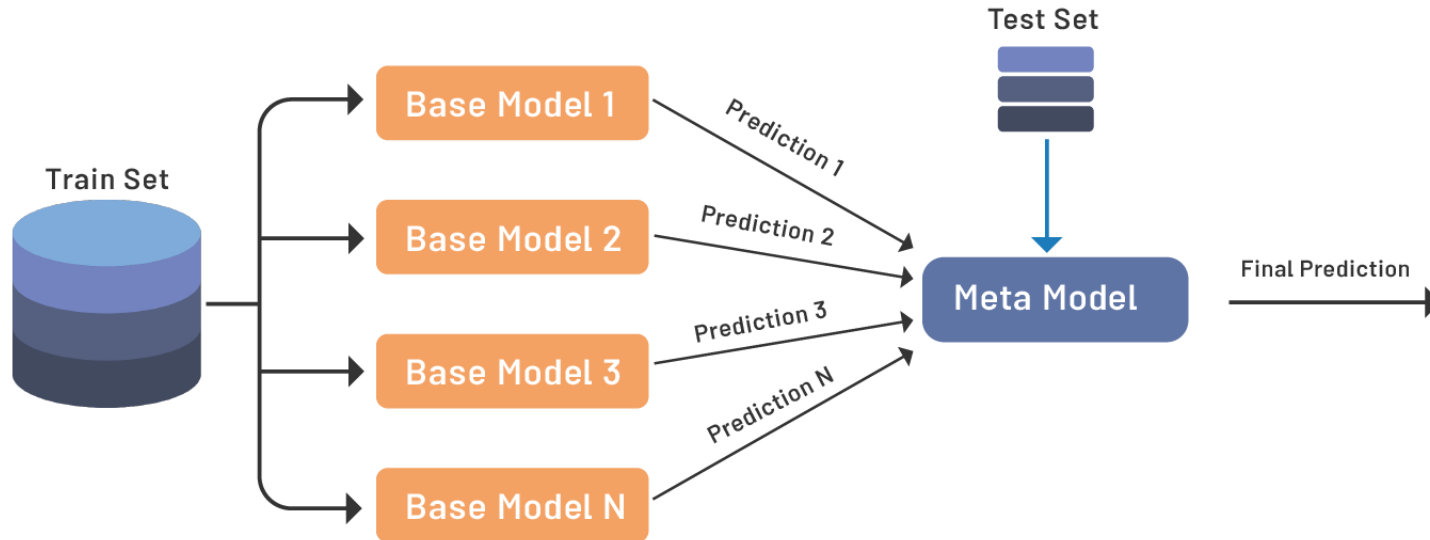
Heterogeneous





Stacking

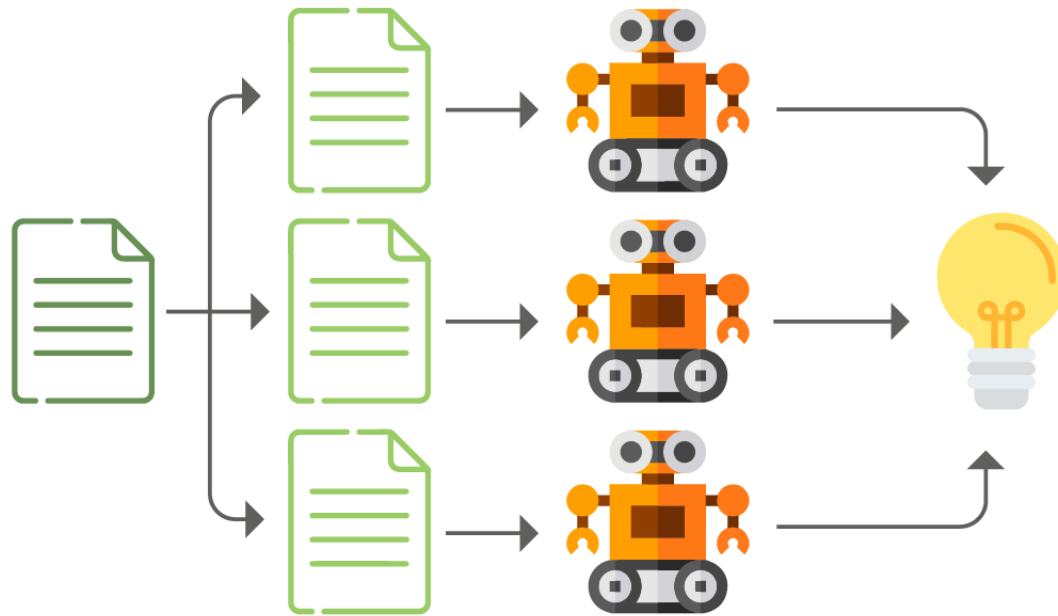
► Employs Heterogeneous base models





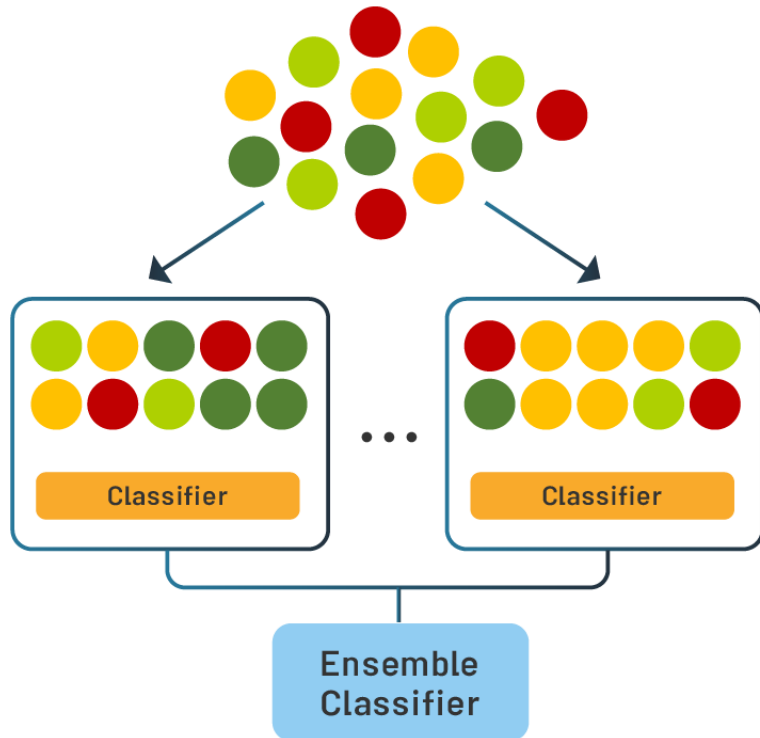
Bagging

- ▶ Bagging trains (homogeneous) base learners in parallel and reduces the variance of estimates by taking the mean of multiple estimates.





Bagging (cont.)



Original Data

Bootstrapping

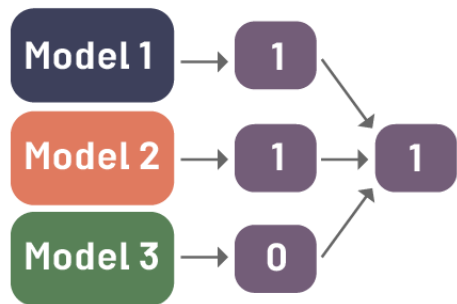
Aggregating

Bagging

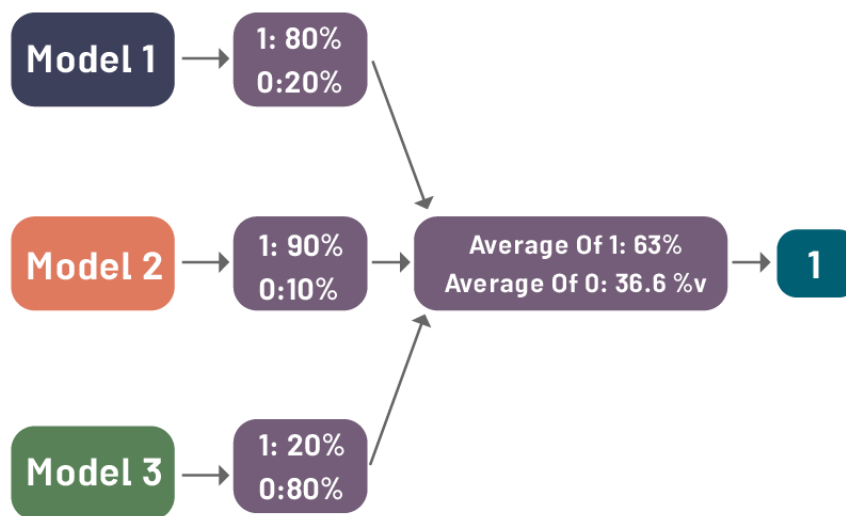


Aggregating Methods

Hard Voting



Soft Voting





Bagging

- ▶ Given m training examples
- ▶ Construct L random samples of size m with replacement (**bootstrap step**)
 - ▶ Each sample called a bootstrap replicate
 - ▶ On average, each replicate contains 63.2% of training data
- ▶ Learn a classifier h_1 for each of the L samples
- ▶ Average over the learned classifiers to produce the final classifier (**aggregation step**)



Bagging (cont.)

- ▶ How Bagging combines classifiers:

$$h: X \rightarrow \{-1, +1\}$$

- ▶ Unweighted voting

$$h(\mathbf{x}_{test}) = \text{sgn}[h_1(\mathbf{x}_{test}) + h_2(\mathbf{x}_{test}) + \dots + h_L(\mathbf{x}_{test})]$$

- ▶ The main idea behind Bagging is to reduce overfitting.
 - ▶ It works well for unstable/ low bias/ high variance models, such as Decision Trees.
 - ▶ It doesn't work for linear models. Why?

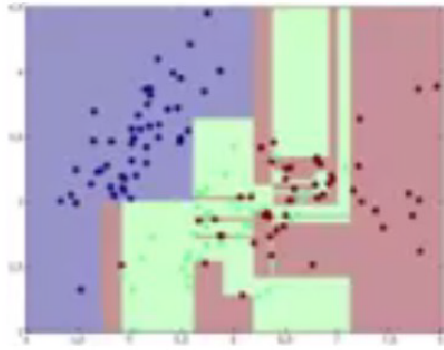


Three Iris Types





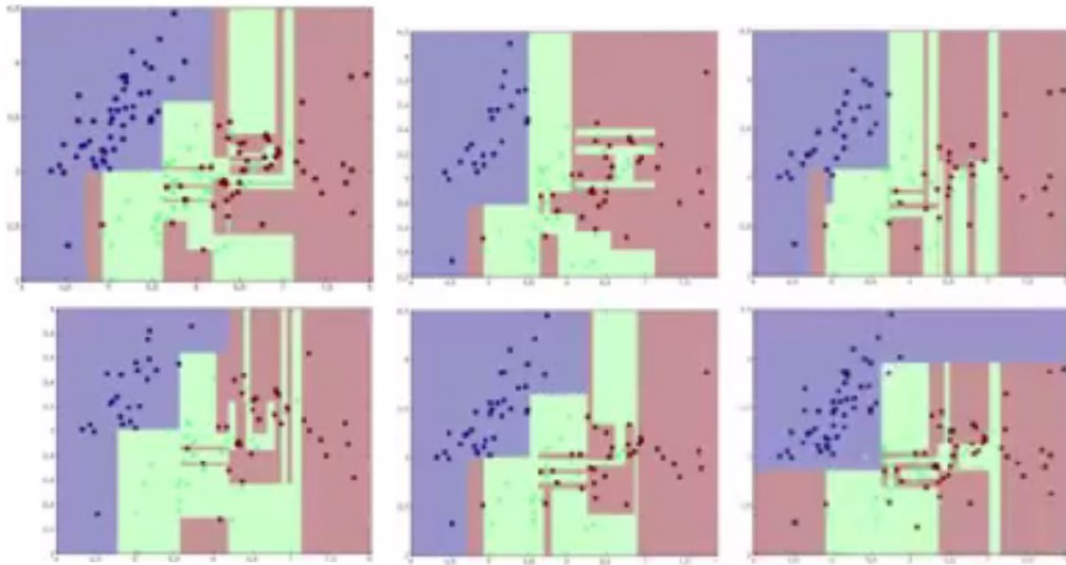
Decision Tree (DT) - decision boundaries



The full Iris.2D dataset



Learned DTs (Bootstrap step)



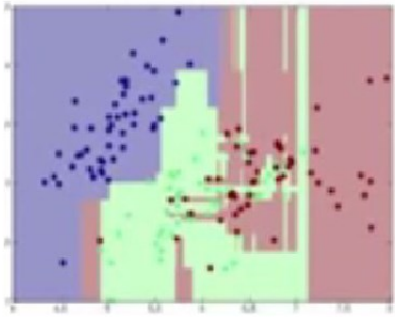


Testing/validation step

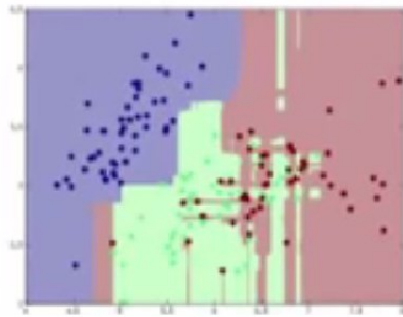
► Aggregating (averaging) over the learned Bagged Trees

► e.g. majority vote/unweighted average (classification)

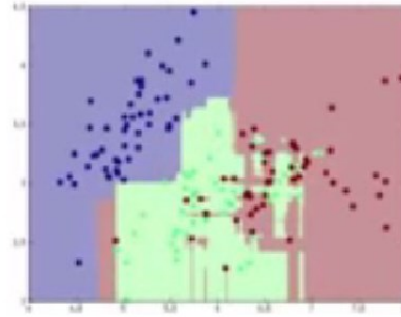
Average of 5 trees



Average of 25 trees



Average of 100 trees



► The Random Forest Algorithm (+ feature bagging)



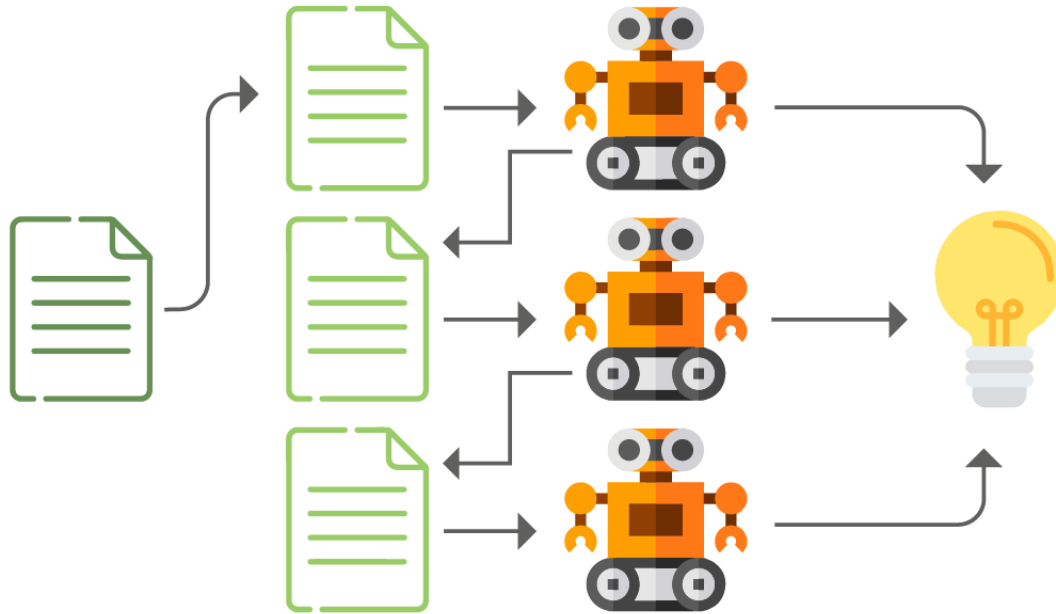
Boosting

- ▶ Boosting is based on the question posed by Michael Kearns (1988) [2]: "Can a set of weak learners create a single strong learner?"
- ▶ Boosting is an ensemble meta-algorithm for primarily reducing bias, and also variance.



Boosting

- ▶ Boosting reduces bias by training (homogeneous) weak learners sequentially, each trying to correct its predecessor.





Adaboost Training Phase

$$\begin{cases} w_i^1 = \frac{1}{m} & \text{Subject to: } i = 1..m \\ \epsilon^1 = \sum_i w_i^1 & \text{Subject to: } i_{\text{th}} \text{ example is incorrectly classified} \end{cases}$$

Total error at step t: $\epsilon^t = \sum_i w_i^t$

Incorrectly
classified examples



Adaboost Final Classifier

► How Adaboost combines classifiers:

$$h: X \rightarrow \{-1, +1\}$$

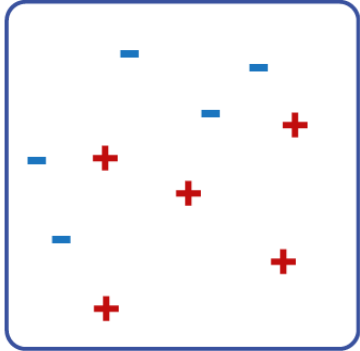
► Weighted voting

$$h(\mathbf{x}_{test}) = \text{sgn}[\alpha^1 h^1(\mathbf{x}_{test}) + \alpha^2 h^2(\mathbf{x}_{test}) + \dots]$$



Adaboost: Components and Steps

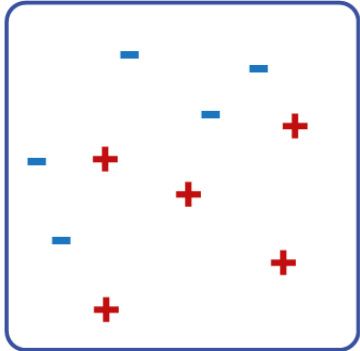
Original data set, D_1



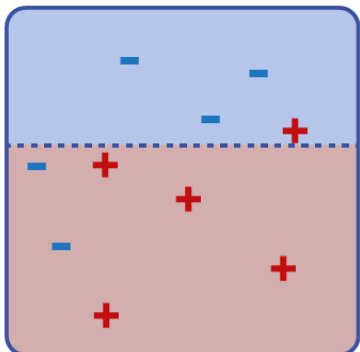


Base Classifier: Decision Stump

Original data set, D_1



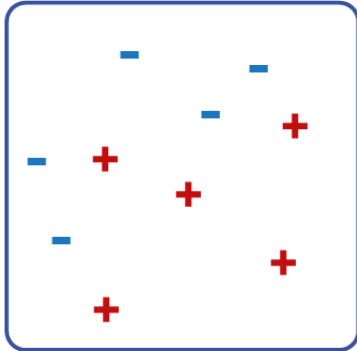
Trained classifier



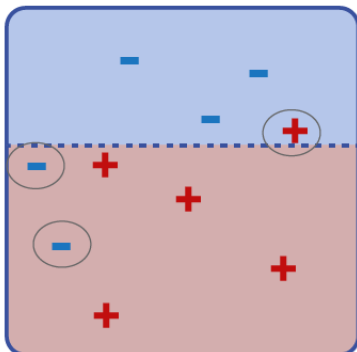


Incorrectly Classified Examples

Original data set, D_1



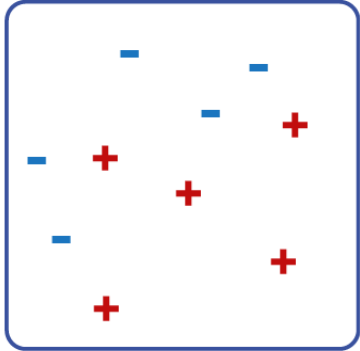
Trained classifier



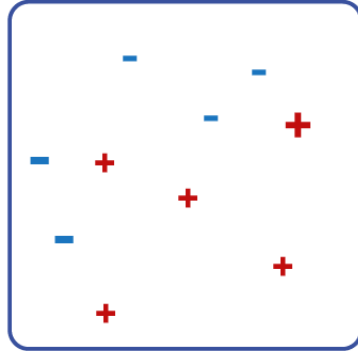


Emphasizing Incorrectly Classified Examples

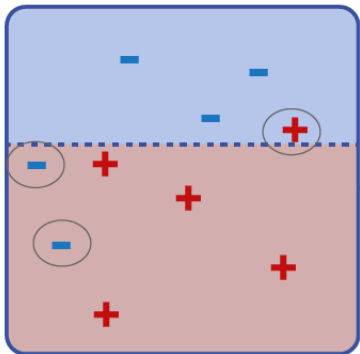
Original data set, D_1



Update weights, D_2



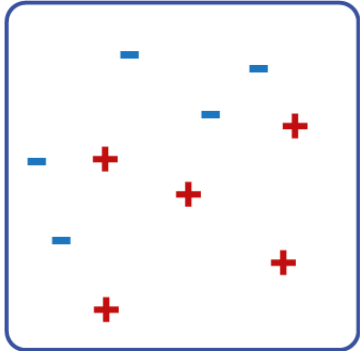
Trained classifier



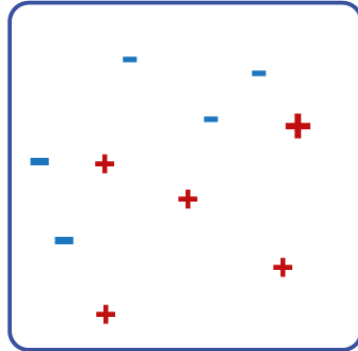


Incorrectly Classified Examples

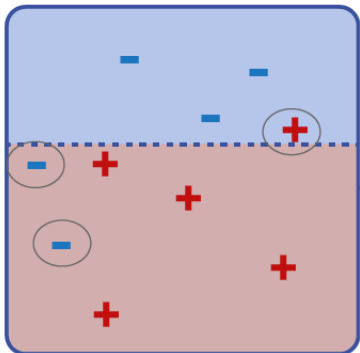
Original data set, D_1



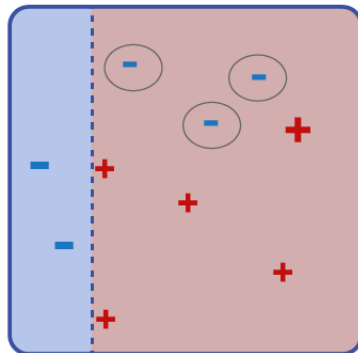
Update weights, D_2



Trained classifier

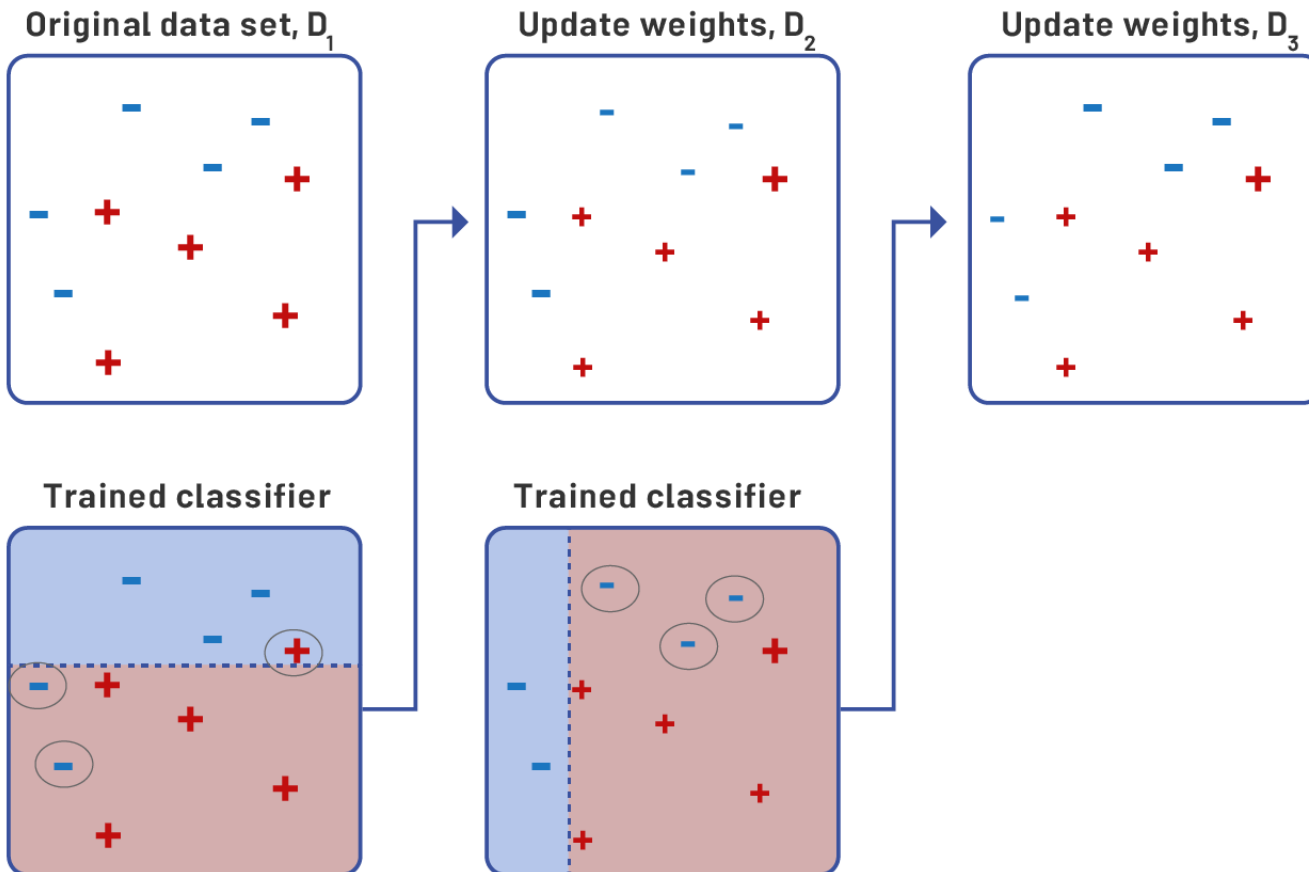


Trained classifier



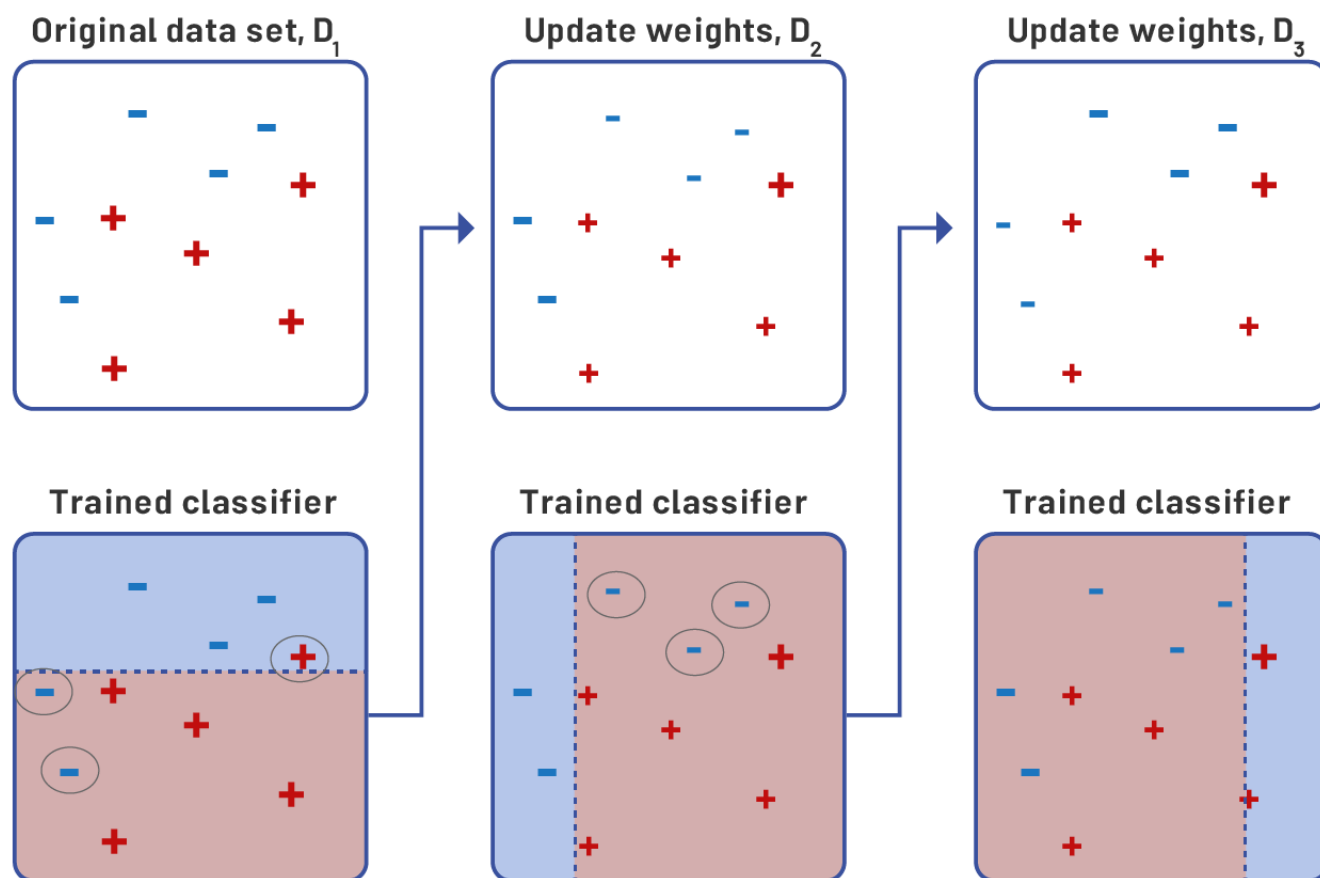


Emphasizing Incorrectly Classified Examples





Training Done

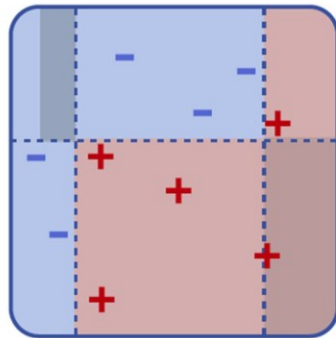




Final Classifier: Weighted Voting

$$.33 * \begin{array}{|c|} \hline \text{Blue} \\ \hline \text{Red} \\ \hline \end{array} + .57 * \begin{array}{|c|} \hline \text{Blue} \\ \hline \text{Red} \\ \hline \end{array} + .42 * \begin{array}{|c|} \hline \text{Blue} \\ \hline \text{Red} \\ \hline \end{array} \geq 0$$

Combined classifier



► Adaboost (Adaptive Boosting)

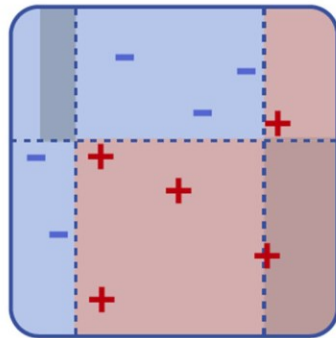
► Yoav Freund and Schapire, 97 [1]



Final Classifier: Weighted Voting

$$.33 * \begin{array}{|c|} \hline \text{Blue} \\ \hline \text{Red} \\ \hline \end{array} + .57 * \begin{array}{|c|} \hline \text{Blue} \\ \hline \text{Red} \\ \hline \end{array} + .42 * \begin{array}{|c|} \hline \text{Blue} \\ \hline \text{Red} \\ \hline \end{array} \geq 0$$

Combined classifier



► AdaBoost (Gödel Prize, 2003)

► This paper [3] (simplified version)



Adaboost Details

► How to update weights at each step:

$$\begin{cases} w_i^{t+1} = \frac{w_i^t}{Z^t} \times \exp[-\alpha^t h^t(\mathbf{x}_i) y(\mathbf{x}_i)] \\ \alpha^t = \frac{1}{2} \ln \frac{1 - \epsilon^t}{\epsilon^t} \end{cases}$$



Adaboost Details (cont.)

► How to update weights at each step:

$$\begin{cases} w_i^{t+1} = \frac{w_i^t}{Z^t} \times \exp[-\alpha^t h^t(\mathbf{x}_i) y(\mathbf{x}_i)] \\ \alpha^t = \frac{1}{2} \ln \frac{1 - \epsilon^t}{\epsilon^t} \end{cases}$$

$$\Rightarrow w_i^{t+1} = \frac{w_i^t}{Z^t} \times \begin{cases} \sqrt{\frac{\epsilon^t}{1 - \epsilon^t}} & h^t(\mathbf{x}_i) = y_i \\ \sqrt{\frac{1 - \epsilon^t}{\epsilon^t}} & h^t(\mathbf{x}_i) \neq y_i \end{cases}$$



Adaboost Details (cont.)

► Normalization factor:

$$Z^t = \sum_i w_i^t \times \sqrt{\frac{\epsilon^t}{1 - \epsilon^t}} + \sum_i w_i^t \times \sqrt{\frac{1 - \epsilon^t}{\epsilon^t}} = 2\sqrt{\epsilon^t(1 - \epsilon^t)}$$

Correctly classified Incorrectly classified



Adaboost Details (cont.)

► Normalization factor:

$$Z^t = \sum_i w_i^t \times \sqrt{\frac{\epsilon^t}{1 - \epsilon^t}} + \sum_i w_i^t \times \sqrt{\frac{1 - \epsilon^t}{\epsilon^t}} = 2\sqrt{\epsilon^t(1 - \epsilon^t)}$$

Correctly classified Incorrectly classified

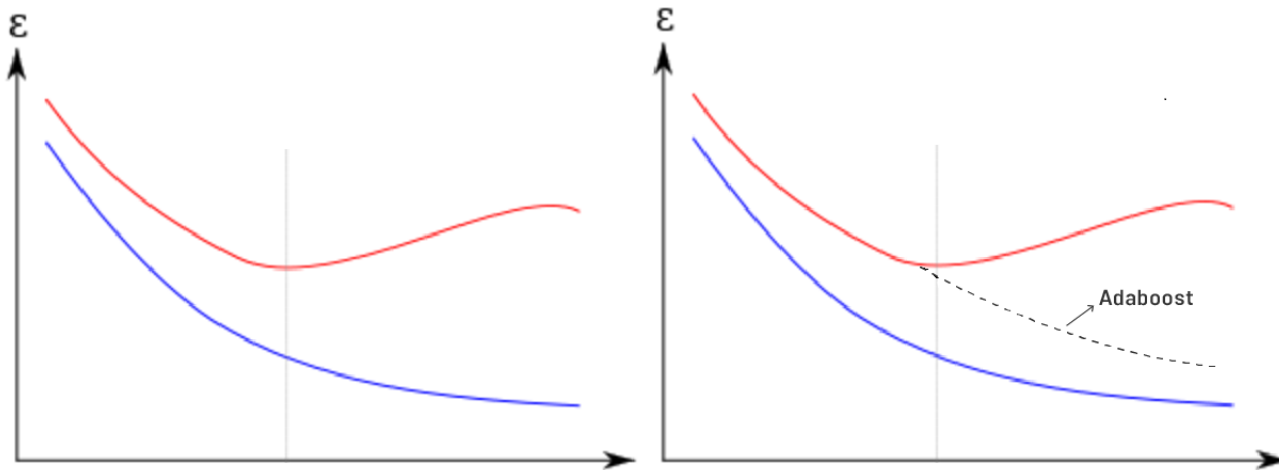
► Weight updating rule:

$$\begin{cases} w_i^{t+1} = \frac{w_i^t}{2} \times \frac{1}{1 - \epsilon^t} & h^t(x_i) = y_i \\ w_i^{t+1} = \frac{w_i^t}{2} \times \frac{1}{\epsilon^t} & h^t(x_i) \neq y_i \end{cases}$$



Adaboost Properties

- ▶ Adaboost is sensitive to noisy data & outliers, less sensitive to overfitting. (Why?)
- ▶ Bias-variance Tradeoff





Adaboost Properties (cont.)

- ▶ Adaboost is sensitive to noisy data & outliers, less sensitive to overfitting.
- ▶ Adaboost (with decision stump/trees as the weak classifier) is often referred to as the best out-of-the-box classifier.



Further Reading

► Other ensemble methods:

► Gradient Boosting

► Extreme Gradient Boosting (XGBoost)

► LightBoost

► CatBoost

► Logit Boost

...

► Random Forest Regression



Further Reading (cont.)

- ▶ T. Dietterich, Ensemble Methods in Machine Learning, International Workshop on Multiple Classifier Systems, Lecture Notes in Computer Science, pp.1-15, 2000.
- ▶ J. H. Friedman, Greedy Function Approximation: A Gradient Boosting Machine, 1999.
- ▶ J. Friedman, T. Hastie, R. Tibshirani, Additive logistic regression: a statistical view of boosting, Annals of Statistics. 28 (2): 337–407, 2000. —→ LogitBoost Algorithm



References

- ▶ 1. Yoav Freund and Robert E. Schapire, A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting, computer and system sciences 55, 119-139, 1997.
- ▶ 2. Michael Kearns, Thoughts on Hypothesis Boosting, Unpublished manuscript (ML class project), 1988.
- ▶ 3. Yoav Freund, Robert E. Schapire, A Short Introduction to Boosting, Journal of Japanese Society for Artificial Intelligence, 14(5):771-780, 1999 (translated).
- ▶ 4. Patrick Winston, Lecture on Boosting, MIT